# Algorithms, 2026 Fall, Homework 1
# (Due: Sep 25)

September 16, 2026

Readings: Our algorithm class assumes basic knowledge about basic algorithms knowledge (like what is a graph, BFS, DFS) and basic data structures (like array, link list, queue and stack), and college mathematics (basic calculus, linear algebra and probability, and big-O notations). In case you have not learnt much before, you are required to read the following chapters.

1. Kleinberg-Tardos book Chapter 2.1-2.4 for big $O$ notations, basic data structures. (You need to finish reading this chapter this week in order to do the homework)

2. Chapter 3 for BFS and DFS, also Chapter 22.1,22.2,22.3 in CLRS. (You need to finish reading this chapter this week in order to do the homework.)

You can discuss homeworks with your classmates or/and using ChatGPT (or other LLMs). But you will have to write down the solution on your own. TA will choose half students to grade each time.

Try to find the solution on your own first before consulting others or LLM. You need to acknowledge other students or LLMs who help you in the homework (specify which problem). If you read some other source that is helpful, you should list all of them.

For those who have extensive experience of OI (show TA the proof), you do not need to solve the problems marked with @.

Problem 1 : @ Write down the pseudocode of BFS (using queues).

Problem 2 : @ KT book pp. 68 problem 5.

Problem 3 : @ KT book pp. 68 problem 6.

Problem 4 : @ KT book pp. 107 problem 2.

Problem 5 : KT book pp. 108. problem 6.

Problem 6 : We are given $2n$ balls and $n$ bins. Each ball is thrown into a random bin (each bin is chosen with probability $1/n$).

Compute the following limit

$$\lim_{n\to+\infty} \mathbb{E}\left[\frac{\text{the number of empty bins}}{n}\right].$$

Problem 7 : For two women $w$ and $w'$, we write $w <_m w'$ to denote that $w$ is worse than $w'$ in the preference list of man $m$. Given two stable matchings $f$ and $f'$ (You can easily construct an example in which there are multiple stable matchings), define a mapping $g = f \vee f'$ as follows:

- for each man $m$, assign him more preferred partner

$$g(m) = f(m) \text{ if } f(m) \geq_m f'(m)$$

$$g(m) = f'(m) \text{ if } f'(m) >_m f(m)$$

- for each woman $w$, assign her less preferred partner

$$g(w) = f(w) \text{ if } f(w) \leq_w f'(w)$$

$$g(w) = f'(w) \text{ if } f'(w) <_w f(w)$$

Show that if both $f$ and $f'$ are stable matchings, so is $g$. (note: We can similarly define $f \wedge f'$. Then, all stable matchings form a distributive lattice, an abstract yet very popular object studied in combinatorics.)